Does Twitter feel the war in real time?

VADER sentiment vs. Russia-Ukraine war events at CHTC scale

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STAT 405 / DSCP, April 2026

Introduction

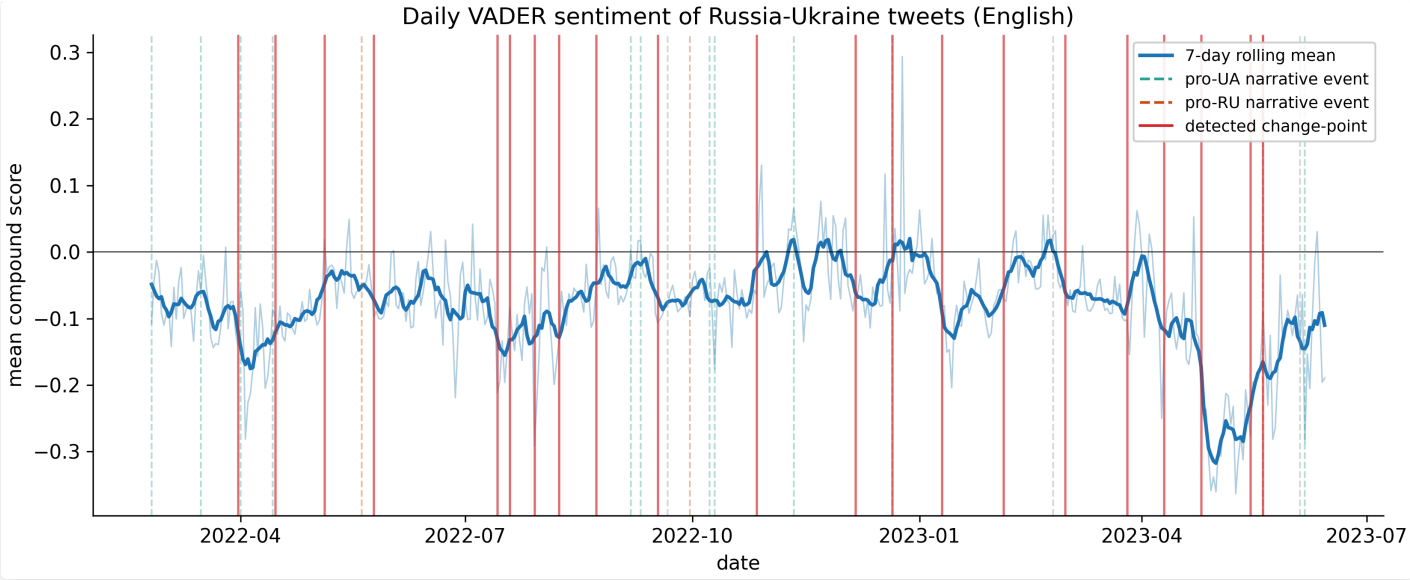
We asked whether the sentiment of public tweets shifts when a major Russia-Ukraine war event happens, how sharply, and how quickly it decays. We scored **70.9 million tweets** (~52 GB on disk) from the Kaggle `bwandowando/ukraine-russian-crisis-twitter-dataset-1-2-m-rows` on CHTC with HTCondor, then ran three analyses against a curated timeline of 21 events: Welch event-study *t*-tests, PELT change-point detection on the daily mean, and an OLS regression of daily sentiment on event dummies with controls. The model explains ~9% of daily variance ($R^2 = 0.092$, $F = 2.44$, $p = 0.0007$); the strongest event effects are **Kherson liberated** (*beta* = +0.12, $p = 0.006$) and **Kakhovka dam** (*beta* = -0.11, $p = 0.019$). Change-point detection flags 21 structural breaks independently, but only one coincides with a labeled event date.

Data, computation, and results

Data. The dataset is ~480 daily CSV files (~52 GB total), Feb 2022 to mid-2024. We used five columns: `text`, `tweetcreatedts`, `language`, `retweetcount`, `followers`. Cleaning was minimal: rows with empty text or an unparseable timestamp were dropped (<2%). After scoring we had 70,876,101 tweets across 476 days. Primary results pool only `language=="en"` because VADER is English-trained; the multilingual pool is reported as a sensitivity check.

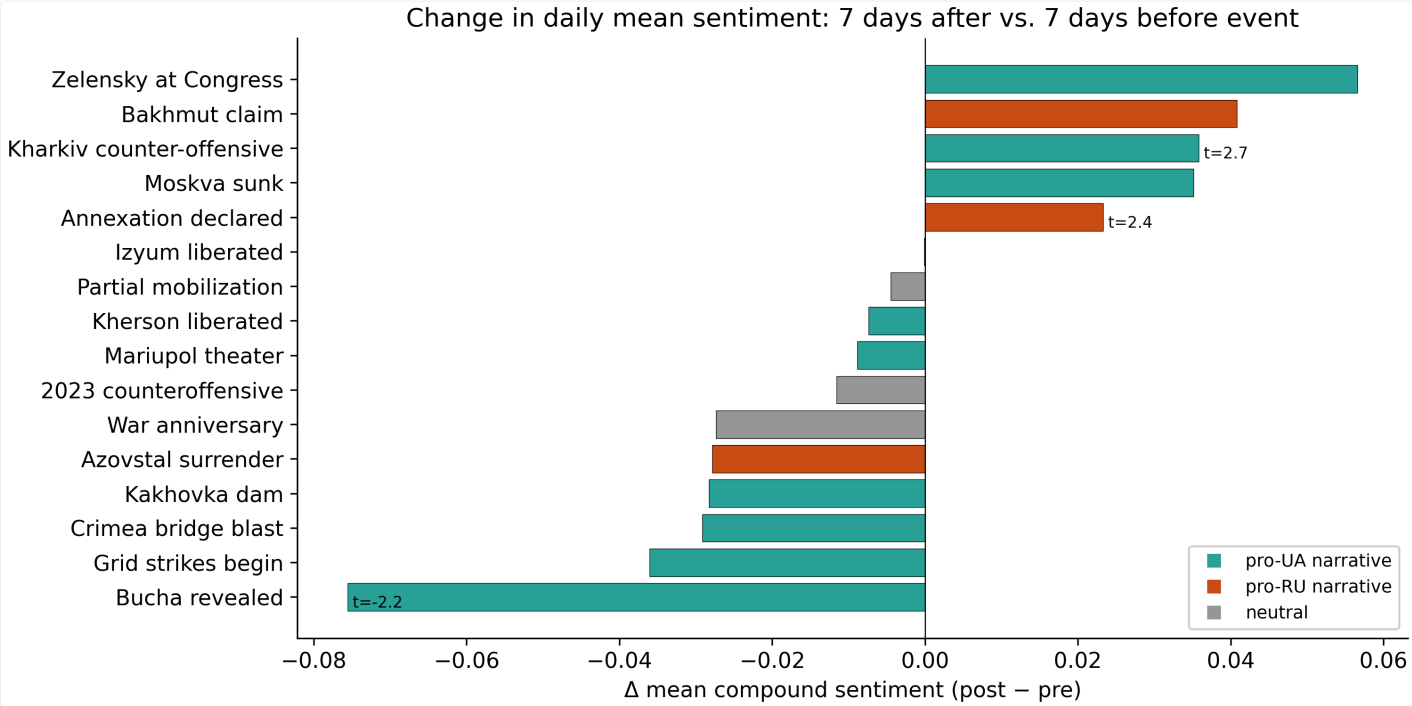
Computation. An Apptainer container (`python:3.11-slim + pandas + vaderSentiment`, ~340 MB) was staged once to `/staging/nomatteson/`. Because `/staging` caps users at 100 inodes, we packed the daily CSVs into 39 multi-day chunks (12 daily files each) before scoring. For each chunk, `score_tweets.py` streams the file in 200k-row blocks, runs VADER per tweet, and emits one row per (date, language) with sums of compound, compound^2, pos/neg/neu, followers, and followers*compound. These are sufficient statistics, so the aggregator never re-reads raw tweets. `aggregate.py` and `analyze.py` then run on the combined output. **Per job:** 1 CPU, peak ~1.2 GB RAM, 8 GB disk. Calibration scored a 419 MB chunk in 7 min; the largest chunks (~3.2 GB) took ~50 min. **Total:** 39 parallel jobs, ~1h40min wall-clock end to end. Raw tweets never left `/staging`.

Results.



Daily VADER sentiment with annotated events and detected change-points

English daily mean compound sentiment drifts around -0.05 to -0.10 for most of the dataset, with the deepest negative excursions around the Bucha revelation (2022-04-01) and the Bakhmut period in spring 2023. Of the 16 events with a full [-7, +7] window, three pass $|t| > 2$ in the event-study test: Kharkiv counteroffensive (+0.036, t = 2.65), annexation declared (+0.023, t = 2.41), and Bucha revealed (-0.076, t = -2.25).



Event study: post - pre daily mean, 7-day windows

The OLS regression (`mean_compound ~ t + log(n_tweets) + event dummies`, n = 476) gives R^2 = 0.092 (adj = 0.054), F = 2.44 (p = 0.0007). The coefficients that clear $|t| > 2$ are Kherson liberated (*beta* = +0.12, p = 0.006), Kakhovka dam (*beta* = -0.11, p = 0.019), Zelensky at Congress (*beta* = +0.089, p = 0.040), and the one-year war anniversary (*beta* = +0.092, p = 0.036). PELT change-point detection flags 21 breaks; only one (2022-12-21) aligns with a labeled

event (Zelensky's congressional address). Reach-weighting by follower count shifts amplitude but not sign across the whole series.

Weaknesses. (i) VADER is English-only, so non-English tweets look falsely neutral; we report the multilingual pool but default to English. (ii) Twitter users are not the public; results describe the platform, not opinion. (iii) Durbin-Watson is 0.70, so daily means are strongly autocorrelated and OLS standard errors are understated. (iv) Events cluster in time, so dummy coefficients are not clean causal effects. (v) No bot filter: coordinated amplification can move daily means without public sentiment actually shifting.

Conclusion

War events explain only a small share of day-to-day variance on English Twitter (R^2 ~ 0.09), but a handful do register statistically: Kherson's liberation and the Kakhovka dam break both cross $|t| > 2$ in the regression with expected sign, and Bucha is the clearest single-event dip in the event study. Change- point detection locates structural breaks, but most don't match our event list, suggesting either that sentiment is driven by things we haven't labeled or that VADER is responding to vocabulary drift rather than events. Future work: multilingual sentiment (XLM-R), an account-activity bot filter, and regression-discontinuity around each event instead of a pooled dummy.

Contributions

Member	Proposal	Coding	Presentation	Report
Calvin Sharpe				
Nils Matteson	1	1	1	1
Pravin Schmidley				
Will Tappa				
Yash Rajani				

Blank cells to be filled per actual contribution after presentation rehearsal; 1 = full, 0.1-0.9 partial, 0 = none.

How to reproduce

```
 git clone https://github.com/matteso1/STAT405project.git
cd STAT405project/chtc
bash setup_on_learn.sh
condor_submit download_and_chunk.sub
condor_submit_dag pipeline.dag
```